

MOHAMED NOMEIR

GENERAL INFORMATION

Country of Citizenship: Egypt
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CAREER OBJECTIVE

I am seeking a research-oriented career in the fields of wireless communication, digital signal processing, information theory & physical layer security, or network modeling that fulfills my personal ambition and enhances my knowledge. At the moment, I am pursuing my Ph.D. at the University of Maryland, College Park. My research focuses on information-theoretic security and privacy in communication systems, and quantum information processing and communications.

EDUCATION AND QUALIFICATIONS

- **Ph.D., student**, Electronics and Communication Engineering Department, **University of Maryland, College Park** January 2022- present
- **M.Sc.**, Electronics and Communication Engineering Department, **American University in Cairo** June 2021
- **B.Sc.**, Electrical Engineering Department, **Alexandria University** June 2018
- **High School Diploma**, **Victoria College**, Alexandria May 2013

PUBLICATIONS

CONFERENCE PAPERS

1. M. Nomeir, A. Aytekin, L. Hu, and S. Ulukus, "Quantum Codes for Private Distributed Matrix Multiplication in the High Privacy Regime", submitted.
2. L. Hu, M. Nomeir, A. Aytekin, and S. Ulukus, "Simultaneously Minimizing Storage and Bandwidth Under Exact Repair With Quantum Entanglement", ITW 2026.
3. M. Nomeir, S. Meel, and S. Ulukus, "Private Information Retrieval With Arbitrary Privacy Requirements for Graph-Based Storage", ITW 2026.
4. S. Meel, M. Nomeir, and S. Ulukus, "Local Private Information Retrieval: A New Privacy Perspective for Graph-Based Replicated Systems", ITW 2026.
5. M. Nomeir, A. Aytekin, L. Hu, and S. Ulukus, "Quantum Codes for Private Distributed Matrix Multiplication for the Low Privacy Regime", ITW 2026.

6. L. Hu, M. Nomeir, A. Aytakin, and S. Ulukus, "Breaking the Storage-Bandwidth Tradeoff in Distributed Storage with Quantum Entanglement", ISIT 2026.
7. M. Nomeir, and S. Ulukus, "Storage-Rate Trade-off in A-XPIR", ISIT 2026.
8. A. Aytakin, M. Nomeir, L. Hu, and S. Ulukus, "Convergence Properties of Good Quantum Codes for Classical Communication", ISIT 2026.
9. M. Nomeir, A. Aytakin, L. Hu, and S. Ulukus, "Quantum GASP Codes for Private Distributed Matrix Multiplication", ICASSP 2026.
10. M. Nomeir, A. Aytakin, and S. Ulukus, "The Capacity of Semantic Private Information Retrieval with Colluding Servers", GLOBECOM 2025.
11. M. Nomeir, P. Dissanayake, S. Meel, S. Dutta and S. Ulukus, "Private Counterfactual Retrieval", Allerton 2025.
12. S. Meel, P. Dissanayake, M. Nomeir, S. Dutta and S. Ulukus, "Private Counterfactual Retrieval With Immutable Features", ISIT 2025.
13. M. Nomeir, A. Aytakin and S. Ulukus, "Byzantine Server and Static Eavesdropper Coalition in a Quantum XTUSPIR System", ICC 2025.
14. L. Hu, M. Nomeir, A. Aytakin, Y. Shi, S. Ulukus and S. Guha, "Entanglement-Assisted Coding for Arbitrary Linear Computations Over a Quantum MAC", ITW 2025.
15. M. Nomeir, A. Aytakin, S. Ulukus, "The Asymptotic Capacity of Byzantine Symmetric Private Information Retrieval and Its Consequences", ISIT 2025.
16. M. Nomeir, A. Aytakin, S. Ulukus, "Quantum X -Secure B -Byzantine T -Colluding Private Information Retrieval", ITW 2024.
17. A. Aytakin, M. Nomeir, S. Ulukus, "Quantum Private Membership Aggregation", ISIT 2024.
18. M. Nomeir, S. Vithana, S. Ulukus, "Asymmetric X -Secure T -Private Information Retrieval: More Databases are not always Better", CISS 2024.
19. M. Nomeir, S. Vithana, S. Ulukus, "Private Membership Aggregation", MILCOM 2023.
20. A. Aytakin, M. Nomeir, S. Vithana, S. Ulukus, "Quantum Symmetric Private Information Retrieval with Secure Storage and Eavesdroppers", GLOBECOM, 2023.
21. M. Nomeir, Y. Gadallah, and K. G. Seddik, "Uplink Scheduling for Mixed Grant-Based eMBB and Grant-Free URLLC Traffic in 5G Networks", WiMob 2021.

JOURNAL PAPERS

1. L. Hu, M. Nomeir, A. Aytakin, and S. Ulukus, "Distributed Storage Systems with Quantum Entanglement Assistance", submitted.
2. M. Nomeir, A. Aytakin, L. Hu, and Sennur Ulukus, "Quantum Private Distributed Matrix Multiplication: New Families of Quantum Codes", submitted.

3. M. Nomeir, A. Aytakin, L. Hu, and Sennur Ulukus, "Quantum Private Distributed Matrix Multiplication: Extending the Classical Codes and Limitations", submitted.
4. S. Meel, M. Nomeir, P. Dissanayake, S. Dutta and S. Ulukus, "What If, But Privately: Private Counterfactual Retrieval", IEEE Transactions on Information Theory, 2026.
5. M. Nomeir, A. Aytakin, S. Ulukus, "Byzantine-Eavesdropper Alliance: How to Achieve Symmetric Privacy in Quantum X -Secure B -Byzantine E -Eavesdropped U -Unresponsive T -Colluding PIR?", IEEE Transaction on Information Theory, 2026.
6. A. Aytakin, M. Nomeir and S. Ulukus, "Quantum X -Secure B -Byzantine T -Colluding Private Information Retrieval", IEEE Transactions on Information Theory, 2026.
7. A. Aytakin, M. Nomeir, S. Vithana, S. Ulukus, "Quantum X -Secure E -Eavesdropped T -Colluding Symmetric Private Information Retrieval", IEEE Transactions on Information Theory, 2025.
8. M. Nomeir, Y. Gadallah, and K. G. Seddik, "Machine Learning-Based Uplink Scheduling Approaches for Mixed Traffic in Cellular Systems", IEEE Access, Jan. 2023.

TEACHING ASSISTANT

AMERICAN UNIVERSITY IN CAIRO (AUC):

- ECNG2105: Circuit Analysis I (Fall 2018)
- ECNG3201: Signals and Systems (Fall 2020-Spring 2020)
- ECNG4302: Fundamentals of Communications II (Spring 2019)
- ECNG4306: Computer Comm Networks (Fall 2020-Spring 2020)
- ECNG4308: Telecommunications Systems (Fall 2020-Fall 2021)

UNIVERSITY OF MARYLAND, COLLEGE PARK (UMD):

- ENEE 222: Elements of Discrete Signal Analysis (Spring 2022-Spring 2023)
- ENEE 662: Convex Optimization (Fall 2023)

RESEARCH ASSISTANT

- **Bridging the Communications Gap: An AI-Based UAV Ubiquitous 5G Communication System** (September 2021 - January 2022)
- **Wireless-Controlled Fully Implantable Smart Micro pumping System for Local Anti-Cancer Drug Delivery in Pediatric Patients** (July 2020 - May 2021)
- **Machine Learning algorithms in 4G systems** (August 2018 - February 2019)

MASTERS THESIS

My thesis, supervised by Prof. Yasser Gadallah and Prof. Karim seddik, is based on network optimization problem in 5G systems; mainly the uplink scheduling for different types of traffic. We analyzed the Grant-Free (GF) user traffic in a hybrid system with Grant-Based (GB) enhanced Mobile Broad-Band (eMBB) users, to reach an optimization problem that can govern the highest data rate for eMBB users while taking in consideration the ultra reliable low latency communications (URLLC) requirements.

B.SC. GRADUATION PROJECT

In our graduation project, supervised by Prof. Said El-khamy, we dealt with problems related to the field of mobile communications security. We dealt with jamming and anti-jamming techniques. We presented different simulation results for A5/1 stream cipher, implemented many enhancements proposed before, and proposed extra enhancements as well. We also worked in physical layer security and implemented co-set codes on LPDC and proposed extra enhancements as well.

FELLOWSHIPS AND AWARDS

- *Medhat Haroun Award for Scholarly Excellence*; Best Engineering thesis in AUC for the theses in 2021.
- *GTA Fellowship and Dean's Fellowship*; University of Maryland.
- *University Fellowship and Laboratory Fellowship*; American University in Cairo.
- *Top 10 Posters Award*; for the poster entitled "GSM System Attacks using Open-BTS Applications", NSRC 2018, Cairo.

REVIEWER

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